

Xiaoyue (Lucy) Liu

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Education

National University of Singapore, Singapore Major in Biomedical Engineering; Minor in Artificial Intelligence and Computing	Aug 2023 – Jun 2027
- Overall GPA: 4.67/5.0 Coursework: Bioengineering Data Analysis, Discrete Structures, Programming Methodology II, Fundamentals Of Biosignals And Bioinstrumentation	
Vanderbilt University, United States, Exchange Student	Dec 2025 – May 2026
- Overall GPA: 3.74/4.0 Coursework: NLP in Bioinformatics , AI in Medical Image Processing	

Research Experience

Undergraduate Research Experience, National University of Singapore	Oct 2024 – Nov 2025
- Developing cardiac digital twin models using multi-view cine MRI data - Processing and harmonizing cross-modal imaging data for 3D heart reconstruction	
Undergraduate Research Intern, Vanderbilt University	Dec 2025 – May 2026
- Developed pipelines for eye CT and MRI analysis, including optic nerve segmentation and evaluation - Assisted in automated anatomical and topological quality assessment for the optic nerve	

Publications

MICCAI 2026, Medical Image Computing and Computer Assisted Intervention Individual First-author paper early accepted (Top 9%) <i>CineMesh4D: Personalized 4D Whole Heart Reconstruction from Sparse Cine MRI</i>	Sep 2026
- Proposed an end-to-end 3D+t whole-heart mesh reconstruction framework from sparse multi-view cine MRI. - Introduced differentiable rendering loss and dual-context temporal modeling for accurate and temporally coherent cardiac reconstruction.	
MICCAI 2025, Digital Twin for Healthcare Workshop Individual First-author oral presentation <i>Personalized 4D Whole Heart Geometry Reconstruction from Cine MRI for Cardiac Digital Twins</i>	Sep 2025
- Proposed the first weakly supervised framework to reconstruct 3D+t heart surface meshes from cine MRI for digital twins. - Accurate reconstruction of whole-heart anatomy and motion enabling automatic extraction of key cardiac indices.	
CVPR 2026 Findings Track Third-author paper accepted <i>Decoupling Frequency for Endoscopic Depth Estimation in Sparsely-Textured Regions</i>	Mar 2026
- Proposed a frequency-decoupled framework for endoscopic monocular depth estimation in sparsely-textured regions. - Improved structural boundary preservation and geometric consistency by rectifying frequency components.	
AAO 2026, American Academy of Ophthalmology Annual Meeting Abstract submitted <i>Automated Orbital Segmentation for Analysis of Muscle Reduction</i>	Apr 2026
- Developed automated orbital segmentation models for CT and MRI to enable standardized volumetric analysis. - Investigated the association between rectus muscle volume reduction.	

Internship

MicroPort Medical (Shanghai) Co., Ltd.

May 2025 – Aug 2025

Research & Development Intern – Cochlear Implants Department

- Developed noise detection and signal processing pipelines for cochlear implant audio signals as the basis for AI-based noise reduction.
- Conducted research on AI-based noise reduction algorithms for cochlear implants to enhance speech perception in noisy environments.

Academic Service

MICCAI 2026, Medical Image Computing and Computer Assisted Intervention

2026

- Served as a reviewer for MICCAI 2026.

Skills

Languages: Python, Java

Technologies: PyTorch, TensorFlow, MATLAB, Computer Vision